

Basic Code Analysis: Root Cause Analysis of Production Outage

Duration : 60 Min.

Scenario

A major production outage occurred during peak hours. Perform a complete RCA by auditing Git commits and application logs. Document findings in 'outage_rca.json'.

Target File Location & Creation

File to Create/Update: student_workspace/outage_rca.json

Task Objectives

Perform the following actions inside the student_workspace directory: - Create/update 'outage_rca.json' inside 'student_workspace/' with fields 'rca_commit_hash', 'trigger_exception', and 'prevention_plan'.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.
2. Initialize the repository history by running: `python setup_git.py`
3. Use Git CLI tools in the terminal to inspect the commit history.
4. Create/update student_workspace/outage_rca.json and record the extracted commit hash and author username.
5. Save your changes (Ctrl + S or Cmd + S).
6. Verify your work by running `python run.py` in the terminal before submitting.

Validation

Once you have saved your files and verified your progress, return to the platform dashboard and click the **"Run Test" / "Verify"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	outage_rca.json exists in student_workspace/ and is valid JSON	5 Marks

TC2	Root cause commit hash identified	5 Marks
TC3	Triggering error exception class identified	5 Marks
TC4	Long-term prevention plan documented	5 Marks

Total Score: 20 Marks

Important Notes

- This is an auto-evaluated question. Ensure all code edits are properly saved and the 'run.py' checks pass before submission.